Latin America and the Caribbean Research and Action Agenda

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Executive summary

Many Small Island Developing States (SIDS) in the Caribbean basin and countries of continental Latin America face economic and environmental challenges associated with climate change.[1] However, climate change is also increasingly recognised as a threat to mental health, compounding risks for poor mental health outcomes and destabilizing the conditions needed for good mental health. While research at the intersection of climate change and mental health has proliferated in recent years, the field remains disconnected, unequal and siloed, slowing urgent progress to address the mental health impacts of climate change and understandings of the mental health benefits of climate action.

Connecting Climate Minds (CCM) is a Wellcome-funded initiative to foster a collaborative, transdisciplinary climate change and mental health field with a clear and aligned vision. Over the last year, we have convened experts across disciplines, sectors and countries to develop regional and global research and action agendas. These agendas set out:1) an inquiry framework for identifying, understanding and addressing the needs of people experiencing the mental health burden of the climate crisis; and 2) priorities for research and translation of evidence into action in policy and practice.

This report presents a proposed research and action agenda for climate change and mental health in Latin America and the Caribbean (LAC).

A scoping review conducted as a foundational step of this project found limited studies within LAC investigating the impact of climate change on mental health. Existing research provides important insights grounded in the cultural and historical contexts of climate and mental health in LAC.[2, 3, 4, 5, 6, 7, 8, 9,10, 11] Yet many studies were conducted after major cyclonic disasters and lack wider consideration of the impact of the climate crisis on mental health. The identified literature mainly focuses on negative mental health outcomes due to acute extreme climatic events in the context of a) the severity of witnessed/experienced trauma/loss, and b) underlying socio-economic vulnerabilities.[2, 3, 4]

Within LAC the risk of experiencing climate-related mental health challenges varies across geographic areas, personal demographics and sociocultural characteristics. However, Indigenous Peoples, ethnic minorities, coastal economy and farming communities, women, children, youth, the elderly, disabled persons and persons with pre-existing mental health challenges appear most vulnerable to climate-related mental health challenges. While more research is needed, traditional medicines and community responses – including by religious groups and those who trust in a power beyond humanity – may be useful tools to address mental health and wellbeing challenges linked to climate change in the region.

Priority research themes for the region cover myriad areas, including:

- Understanding and defining concepts of self-perceived climate-related mental health vulnerability and resilience;
- Defining key diagnostic terms such as eco-anxiety;
- Quantifying the ‘additional’ mental health burden associated with the climate crisis;
- The disruptions in mental healthcare pathways occurring secondary to the climate crisis; and
- Understanding the pathways and mechanisms by which the climate crisis is affecting mental health in LAC, including:
    - The extent to which climate crisis-related modulations of general socioeconomic or human development indicators give rise to mental health challenges or mental wellbeing distortions;

- The impact of heat on mental health outcomes, including depressive states and incidence of self-harm or death by suicide;
- Climate-related circadian rhythm distortion due to changes in living and sleeping conditions, and accompanying incidence of poor mental wellbeing or effects on mental health in children and Indigenous Peoples (these were the groups identified as being vulnerable to this issue, but it is likely that it is far more widespread); and
- Climate-related cortisol changes and the occurrence of stress.

Operationalising a research agenda in LAC will require respectful, contextually relevant and culturally appropriate research. Participatory approaches to research will be crucial, with involvement of Indigenous Peoples, youth and civil society organisations (especially where formal national governance structures are fragile). Engagement with lay community members as co-creators, implementers and disseminators of research will be important to ensure a sense of belonging, agency and purpose for climate-related mental health action, especially among youth.

We hope this agenda will help to focus the efforts of funding, policy and practice communities working toward a future where no one is held back by climate-related mental health challenges.

Quick reference: Key takeaways

1. The impact of climate events on mental health and wellbeing in LAC may be compounded by political, cultural, social and economic conditions.
2. Risk varies by geography, demographics and sociocultural characteristics. Indigenous Peoples, ethnic minorities, coastal economy and farming communities, women, children, youth, the elderly, disabled persons and those with pre-existing mental health challenges were perceived as being most vulnerable to climate-related mental health and wellbeing disruptions and possible new onset mental health challenges.
3. Adverse climate events may trigger events leading to socioeconomic challenges that negatively impact mental wellbeing (e.g., drought can lead to loss of food security, which can lead to forced migration and displacement, which can lead to poverty related vulnerabilities, which can lead to distortion in mental wellbeing). It is also possible that interdependent and cascading biological social and psychological factors secondary to exposure to adverse climate events may exacerbate existing mental health challenges. These disruptions may arise due to alterations in care pathways.
4. Identifying population sub-groups who experience increased vulnerability to mental health challenges caused by adverse climate events, and conversely those experiencing resilience to these effects is equally important. Therefore, there is a need for bidirectional research. This work will require the use of mixed methods and should seek to establish definitions and clinical criteria for mental health conditions related to climate change (e.g., eco-anxiety) and should also seek to validate exiting clinical tools within LAC. The progress of research in the region will be impacted by the current lack of mental health surveillance systems; therefore, research should include human (e.g., workforce) and infrastructural capacity building elements.
5. Cultural awareness and appropriateness are essential to successfully conducting climate and mental health research in LAC. Researchers should note that Indigenous Peoples in LAC are a diverse group; not all Indigenous Peoples have a spiritual world view.
6. Collaborative research (multisectoral and interdisciplinary) will be valuable in resource-limited settings within LAC.
7. Partnerships between research funders and civil society in LAC may be useful where formal national governance structures are fragile.
8. Engagement with youth as co-creators, implementers and disseminators of research is critical.
9. Community created responses may generate useful solutions to addressing mental wellbeing and mental health challenges occurring secondary to climate change.
10. Resilience and belonging through community advocacy, agency and purpose are possible co-benefits of climate action; researchers should investigate methods to monitor and evaluate these co-benefits.

Introduction

Context

Climate change and mental health are two of our greatest global challenges, and awareness of the intersection between mental health and the climate crisis has grown rapidly in recent years.[12] Climate change exacerbates mental health challenges by increasing exposure to extreme heat and the traumas of extreme weather events,[13] destabilising the conditions needed for good mental health and wellbeing (for example, water and food insecurity, forced migration, polluted air, loss of treasured environments),[13] disrupting access to healthcare[14], and increasing psychological distress through awareness of climate threats and insufficient climate action.[15] People living with mental health challenges are also particularly vulnerable to the stressors of the climate crisis, such as increased risk of physical heat stress and death during heatwaves.[16, 17, 18]

In response to the mounting mental health toll of the climate crisis, research in the climate and mental health field has grown rapidly. Nevertheless, key evidence gaps exist in many regions, including the mental health burden attributable to climate change, the pathways and mechanisms underlying these impacts, the co-benefits of climate action for mental health and the best interventions or solutions to support mental health in a changing climate. Climate change and mental health research remains frustratingly disconnected across disciplines, sectors, and geographies, and is unevenly focused on certain topics and global regions.[19] Moreover, siloed decision making slows the translation of evidence to aligned action across climate and mental health policy and practice.[20, 21] A more inclusive, connected agenda is urgently needed to generate the evidence to truly understand, monitor and respond to the interconnections between climate change and mental health.

Connecting Climate Minds

Connecting Climate Minds (CCM) is a Wellcome-funded project launched in 2023 to develop an inclusive agenda for research and action in climate change and mental health. The project has two key, intertwined aims. The first is to develop an aligned and inclusive agenda for research and action that is grounded in the needs of those with lived experience of mental health challenges in the context of climate change, to guide the field over the coming years. The second is to kickstart the development of connected communities of practice for climate change and mental health in seven global regions (designated by the Sustainable Development Goals), equipped to enact this agenda. We aim to combine the strengths of a global perspective and regional focus, and bring together diverse disciplinary perspectives into a shared vision that can ensure research is effective at addressing priority evidence gaps and informing changes in policy and practice at the intersection of climate change and mental health.

Through bringing together experts across diverse disciplines, sectors and countries, the CCM team has facilitated the development of a lived experience informed research and action agenda for the climate change and mental health field in Latin America and the Caribbean (LAC).

Objectives of the research and action agenda

The research and action agenda is designed to focus future efforts to help those who are experiencing, or will experience, the compounding mental health challenges of climate change. It aims to support those who are already responding to these challenges – through communities, research, policy and practice – by building a more connected and collaborative climate change and mental health field. It also aims to empower experts across disciplines and sectors to join and make progress in this area by identifying clear priorities and fostering a more inclusive and transdisciplinary field.

The agenda addresses these aims through three core objectives, which are to:

1. Identify priorities for research that can inform action to meet the needs of people experiencing and responding to the mental health impacts of climate change in sub-Saharan Africa.

2. Identify what is needed to appropriately conduct research and translate evidence to action in policy and practice in sub-Saharan Africa.

3. Build understanding among researchers, funders, and policy experts across disciplines and sectors of their role in furthering climate change and mental health research, and equip them with these clear and actionable priorities.

The regional agenda will be integrated with six other regional agendas to inform a **global research and action agenda** for climate change and mental health. This will ensure that global research efforts and investment in climate change and mental health are grounded in regional-level priorities. Importantly, the global agenda will also integrate insights from agendas developed with and for some of the most affected groups globally, namely Indigenous Peoples, youth, and small farmers and fisher peoples.

Use of the terms climate change and mental health

Climate change, mental health and their intersection are complex and wide-ranging fields. For the purpose of this agenda, we define the scope of these terms as follows.

By **mental health challenges**, we mean thoughts, feelings, and behaviours that affect a person's ability to function in one or more areas of life and often involve significant levels of psychological distress. This includes, but is not limited to, anxiety, depression, post-traumatic stress, psychosis, suicidal thoughts and substance misuse.

By **experiences of the effects of climate change**, we mean: 1) experiencing direct impact of climate hazards, such as more frequent and intense heatwaves, wildfires/bushfires, drought, floods or storms (e.g., typhoons, hurricanes, cyclones), and 2) experiencing disruption to the social and environmental determinants of good mental health, such as being forced to move home, not being able to access food or water, losing livelihood or homelands, or disruption to cultural practices because of climate change.

Mental health challenges in the context of climate change include:

- How climate change may lead to worsening pre-existing mental health challenges.

- How climate change may contribute to prevalence or impact of existing mental health challenges.

- How climate change may impact treatment access or effectiveness for those with mental health challenges.

- How climate change may lead to new mental health challenges.

The convening work of CCM presents a key opportunity to build our understanding of diverse perspectives, framings and terminologies in LAC, which we have also sought to reflect within the research and action agenda.

Background to Connecting Climate Minds

Regional Community of Practice

The CCM project in LAC is led by a Regional Community Team (RCT), responsible for convening diverse expertise across the region and building regional capacity to create and enact the research and action agenda. The structure of the RCT is outlined below.

Regional Community Convenor (RCC)

Purpose: Responsible for developing and delivering project activities in the region, including convening and supporting a regional community of diverse expertise.

Members:
Project Team Leader: **Ambassador Gillian Bristol**, Latin American Caribbean Centre (LACC), Vice-Chancellery, The UWI Mona, Jamaica
Lead Researcher/Project Manager: **Dr. Sandeep Maharaj**, The UWI St. Augustine Campus,
Technical Advisor: **Dr. Natalie Greaves**, The UWI Cave Hill
Assistant Researcher: **Dr. Shamjeet Singh**, The UWI St. Augustine
Assistant Researcher: **Dr. Satish Jankie**, The UWI St. Augustine
Communications /Administration: **Mrs. Lina Torres Maestre**
Coordinator: **Mrs. Chandika Ganesh**
Assistant: **Mrs. Jacinth Seemungal**

Co-convenors

Purpose: Bringing additional breadth of expertise across disciplines and countries (i.e. organisations spanning climate expertise, stress neuroscience and mental health expertise, in different sectors), providing technical advice and review, and supporting project delivery.

Members:

Professor Enrique Falceto De Barros, Director of WONCA (World Organization of Family Doctors). Programa de Pós-Graduação em Educação em Ciências (PPGECI) at the Universidade Federal do Rio Grande do Sul (UFRGS)

Mr. Marcos Williamson, Educational Investigator and Director of the Environmental Information Centre of the Universidad de las Regiones Autónomas de la Costa Caribe Nicaragüense (URACCAN), Nicaragua

Dra. Martha Rosa Munoz, Director of FLACSO Cuba-The Latin American Faculty of Social Sciences (FLACSO), Cuba

Dr. Fresia Hernandez, National Director of the Clinical and Health Psychology Programme at the Tecnologico de Monterrey (TEC), Mexico

Dr. Dassaëve Brice, EarthMedic and EarthNurse Haiti

Dr. Clemencia Ramirez, Independent Consultant, International Migration Organisation (IMO)**.**

Assistant Co-Convenors/Dialogue Facilitators:

Dr. Tatiana Camargo, Brazilian Planetary Health Hub at Universidade de Sao Paulo and Federal University of Rio Grande do Sul.

Dr. Raquel Santiago, Planetary Health (PH) Brazil, IEA/USP

Professor Jonathan Sherin, The Healthy Brains Global Initiative

Lived experience advisory group (LEAG)

Purpose: Advisory board of experts with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. Drawing on their unique expertise and wisdom, LEAGs provide vital community-centred perspectives and guidance that inform the overarching approach and outputs of the project.

Members:

Ms. Vashti Burrows, Bahamas

Ms. Gloria Blaise, Haiti

Ms. Raquel Tupinamba,Tupinamba Indigenous Council of Baixo Tapajos/Amazon,

Mr. Norton Chavarria, Karata Community, Nicaragua

Youth ambassador (YAs)

Purpose: Youth advisors (aged 18-29) with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. YAs bring unique youth-centred perspectives to the development and implementation of project activities.

Members:

Ms. Ana Gabriela Mejia - Ecuador

Mr. Harvey Vijay Sharma - Guyana

Methodology

- We produced this research and action agenda through a robust and inclusive methodology to capture, combine and refine a rich diversity of perspectives while fostering connection across a growing community of practice. Given the need to generate descriptive lived experience data on a relatively novel and sensitive topic, a pragmatic mixed-methods approach was used leaning heavily on principles of qualitative research design.

- The CCM core team developed this methodology in consultation with Regional Communities of Practice, a Global Advisory Board and Wellcome, which facilitated important iterative changes to methodology. Methods and materials were adapted regionally to ensure a balance of global standardisation with regional appropriateness and flexibility. Continuous sharing between regions of processes, learnings and challenges was critical to the iterative development of the methodology. The overarching process for developing the regional research and action agendas is shown below.

Figure 1: Research and action agenda development methodology

Pre-dialogue scoping

Global scoping and framing were conducted prior to the conduct of research in the SDG regions in the CCM project. During global scoping the CCM core team performed:

- A global-level scoping review of current reviews, key papers and policy reports relating to the climate change and mental health field;
- A mapping of the research categories covered by and used to structure previous reviews of the field; and
- A second mapping of the recommendations for action on climate change and mental health proposed by reports written to inform policy and practice.

They then used these results to frame the line of inquiry for the dialogues and research agenda to ensure alignment with the current field while responding to key gaps.

Scoping research conducted with LAC

- In addition to the above, LAC's RCC project team leader and lead researcher engaged with lived experience experts and specialist advisors in the field through key informant discussions to ensure that the production of the research agenda was appropriately grounded in an understanding of the key lived experience needs and emerging policies in the region.
- LAC's RCoP also conducted a scoping review to summarize the evidence related to climate events/threats and associated mental health outcomes, including underlying mechanisms specific to the region.
- The reviewed literature identified gaps in knowledge and contextual information regarding climate and mental health in the LAC, (these are provided in Appendix 1, while the methodology of the review, PRISMA diagrams and summary of the reviewed papers are presented in Appendix 2).
- LAC's team also distributed an **online survey** prior to Dialogue 1 (**pre-dialogue survey**) to solicit perceptions on regional climate impacts, climate-related mental health impacts, and research priorities.
- The results of all global and regional specific scoping activities were then combined to finalise the design of regionally relevant and culturally appropriate dialogues.

Dialogue methodology

- Experts across disciplines and sectors were purposively sampled and recruited by snowballing[i] through regional co-convenor networks to attend two virtual dialogues (3 hours each).
- The framework for the dialogues was designed by the global team and adjusted by the RCoP-LAC to increase cultural appropriateness and ease of use across the diverse languages and technological capacities in the region (see Appendix 3 for dialogue agendas).
- The dialogues were facilitated by a dialogue unit comprising members of the RCoP-LAC, who are themselves researchers, policy makers and lived experience experts, along with colleagues from the CCM core team and Red Cross Red Crescent Climate Centre.
- Members of the dialogue unit presented in plenary sessions and served as discussion facilitators, notetakers or non-participating observer researchers in break out rooms, thereby contributing to data production, collection and analysis.
- The first dialogue identified climate/mental health hazards and risks, mapped vulnerable populations and gathered insights into perceived and experienced regional needs and research priorities.
- The second dialogue gathered feedback on research priorities (from Dialogue 1) and identified thoughts and recommendations on conducting research and translating evidence into policy and practice within LAC. It also explored the diversity of regional perspectives and understandings of key relevant concepts.
- Data generated in the dialogues included: Zoom chats, notes made by dedicated notetakers, transcripts of breakout room/ plenary discussions and Google Jamboard notes written by participants (Dialogue 2 only).

[i] Snowballing refers to potential participants identifying and referring other suitable participants.

Survey methodology

- We distributed a **post-dialogue online survey** after Dialogue 2 to check findings, obtain a second round of feedback on research priorities and identify any relevant methods, metrics and datasets to address these priorities.

Analysis methodology

- **Research categories:** As previously described, the Climate Cares Centre conducted a global landscaping exercise of relevant existing climate change and mental health reviews and identified four broad research categories as areas of critical need for further work globally. This framework was used as the basis for structuring discussions within dialogues to generate research priorities and formed the global coding framework for analysis.
- **Generating priority research themes**: Participants in Dialogue 1 were guided through a semi-structured discussion exploring their views on the emerging and likely mental health consequences of current and future regionally relevant climate hazards and opportunities for mental health benefits of action. Participants then conceptualized possible research themes based on where evidence could usefully inform/impact policy and practice (see Appendix 4 for criteria for selection of priority research themes).
- **Analysis:** Dialogue data (transcripts of breakout rooms and notes) was analysed using a hybrid thematic framework method – a matrix-based approach that allows qualitative researchers to undertake deep interrogation of data (e.g., transcripts and written notes) to aid in the identification of recurring patterns in the data.
    - The analyst team (see Appendix 5 for description) used a coding framework to code dialogue transcripts (see Appendix 6).
    - Nuanced summaries of the identified patterns were then produced in alignment to the four broad categories identified from the global landscaping exercise and summarised in a table matrix.
    - Key quotes drawing from the notes and transcripts of the breakout discussions are presented further in this report as contextual evidence to support the themes.
- **Draft priority research themes:** The nuanced summaries noted above were used to draft a list of priority research themes based on triangulation across breakout notes, transcripts, pre-dialogue scoping and expert consultation.
- **Refinement of priority research themes:** Research themes (in the form of questions) were shared with Dialogue 2 participants. who gave feedback in response to the following prompts: *'Is this an important research topic for LAC? Should anything be added or taken away from it?'* (The prompts were presented in lay language form to increase accessibility to all participants). Research themes were then refined in response to this feedback and shared with dialogue participants for comments in a post dialogue survey as a form of participant checking. In the survey introduction, participants were thanked (culturally necessary) and then invited to make needed changes. The post-dialogue survey was also circulated to a wider sample of experts who were not participants in the dialogue with the prompt: *Is there anything you would like to change, remove or add to this research theme?'* All participants in the post-dialogue survey were given the option to suggest additional research themes.

- **Finalisation of priority research themes:** The final list of priority research themes generated incorporated survey feedback, consultation with the RCoP-LAC, regional experts, CCM core team, Global Advisory Board and Wellcome.

Participants

Dialogue participants were a diverse group across geographical spread, gender, sector, and discipline. All participants were invited to both dialogues, however in some cases participants were unable to attend both dialogues.

In total 45 participants attended Dialogue 1 and 30 participants attended Dialogue 2. The tables below provide a breakdown of participant characteristics.

Geographical spread

	Dialogue 1		Dialogue 2	
Country	Number	Percentage	Number	Percentage
Bahamas	1	2.5%	1	3.80%
Barbados	1	2.5%	1	3.80%
Bolivia (Plurinational State of)	1	2.5%	0	0%
Brazil	4	11%	5	19%
Chile	2	5%	0	0%
Costa Rica	1	2.5%	1	3.80%
Ecuador	1	2.5%	0	0%
Grenada	2	5%	0	0%
Guatemala	1	2.5%	1	3.80%
Haiti	4	11%	5	19%
Honduras	3	8%	0	0%
Jamaica	3	8%	2	8%
Mexico	3	8%	1	3.80%
Nicaragua	3	8%	2	8%
Saint Lucia	1	2.5%	0	0%
Trinidad and Tobago	5	13%	7	27%

United States of America	2	5%	0	0%

Expertise

	Dialogue 1		Dialogue 2	
Expertise	Number	Percentage	Number	Percentage
Climate Change	12	21%	7	19%
Mental Health	13	23%	11	29.7%
Health	18	31.5%	14	37.8%
Other	14	24.5%	5	13.5%
I do not know / Prefer not to say	0	0%	0	0%

Discipline

	Dialogue 1		Dialogue 2	
Discipline	Number	Percentage	Number	Percentage
Activism	5	4.5%	4	5%
Community	13	12%	9	11%
Education	21	19%	17	21%
Expert through my own lived experience	5	4.5%	4	5%
Funding	1	1%	0	0%
Healthcare	16	14.5%	12	15%
Non-governmental Organisation	13	12%	9	11%
Policy	4	3.5%	4	5%
Research	25	23%	19	24%
Other	7	6%	2	3%

Gender

	Dialogue 1		Dialogue 2	
Expertise	Number	Percentage	Number	Percentage
Men	8	36%	12	46%
Women	13	59%	14	54%
Non-Binary	1	5%	0	0%
I do not know / Prefer not to say	8	36%	12	46%

Survey participants:[ii]

Pre-dialogue survey: 76

Post-dialogue survey: 6

Ethics, data collection and storage

This study has been reviewed and given an ethically favourable opinion by the Imperial College Research Ethics Committee (Study title: 'Global Dialogues to set an actionable research agenda and build a community of practice in climate change and mental health', study ID number: 6522690). In addition, the study protocol with necessary local adaptations was reviewed and approved by the Institutional Review Board of The University of the West Indies, St Augustine [CREC-SA. 2298/08/2023].

Details on data collection and storage can be found in Appendix 7.

Current state and emerging needs for climate change and mental health in Latin America and the Caribbean

As the climate crisis escalates, more people globally are experiencing related mental health consequences. However, the current evidence base doesn't fully capture these experiences. To develop research themes that ultimately meet the needs of those experiencing and responding to the interconnections between climate change and mental health, it is vital to know: 1) what do people from different backgrounds, contexts and sectors – particularly those with lived experiences of mental health challenges in the context of the climate crisis – report as their experiences, needs and resiliencies, and

[ii] Please note numbers are approximate and do not account for duplicate or incomplete responses.

2) what evidence do people making decisions and taking actions on the ground need in order to adequately respond?

This section sets out the context of the research agenda, exploring the climate hazards facing the region, the current state of evidence on mental health and climate change in LAC, and key insights and emerging needs from the region.

Climate hazards in LAC[iii]

Latin America and the Caribbean (LAC) is facing an increase in the frequency and severity of a range of climate hazards, including the following, modelled to approximately 2030 as compared to historical baseline (generally 1986-2005): [22, 23, iv]

- **Extreme temperatures,** particularly in the tropical regions of South America, Central America and Cuba and the coastal regions of Mexico (with variation across climate models; medium confidence); [v]
- **Droughts and wildfires** in North Argentina, the more populous regions of Brazil and areas of Mexico (high confidence);
- **Heavy precipitation events and flooding,** with the intensity and frequency of extreme precipitation and pluvial floods projected to increase in South America (medium to high confidence), though Central America and the Caribbean Islands are more likely to experience a decline in the average rainfall (low confidence);
- **Sea level rise,** contributing to increased coastal flooding in low-lying areas and shoreline retreat along most sandy coasts, particularly around Central and South America (high confidence); and
- **Tropical cyclones,** with a projected median increase of around 3%.

Climate change and mental health in Latin America and the Caribbean: A summary of the scoping review

A scoping review conducted as a foundational step of this project found limited studies within LAC investigating the impact of climate change on mental health. Existing research provides important insights grounded in the cultural and historical contexts of climate and mental health in LAC.[2-11] Yet many studies were conducted after major cyclonic disasters and lack wider consideration of the impact of the climate crisis on mental health. The identified literature mainly focuses on negative mental health outcomes due to acute extreme climatic events in the context of a) the severity of witnessed/experienced trauma/loss, and b) underlying socio-economic vulnerabilities.[2, 3, 4] Therefore, there is a need for a research and action agenda which explores the full spectrum of acute and creeping climate events impacting the region. The positive influence of green spaces similarly merits further attention.[8]

[iii] Information based on Red Cross Red Crescent Climate Centre mapping exercise and projections (methods detailed in Appendix 8).

[iv] Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the CMIP6 multi-model ensemble provided in the IPCC, 2021.

[v] Within the IPCC and other major sources of climate projections, confidence levels are given on a scale of low, medium, high. The ranking refers to the robustness of the evidence available and the agreement between climate models.

The scoping review further suggested that a research action agenda for LAC should be discussed and investigated in concert with migration patterns and other socio-political and economic factors that may influence or mediate the relationship between climate change and mental health.

Summary of results from dialogues and surveys

This subsection presents findings from the previously described dialogues and surveys relating to perceptions of four overarching research categories, as listed below. These categories were identified as areas of critical need for further work globally and that map the climate and mental health research space at a high level, based on an initial review of the literature. The priority research themes in the following section are also arranged within these four categories.

- **Impacts, risks and vulnerable groups**: improving our understanding of the extent to which mental health is affected by climate change and for whom. For example: what mental health outcomes are impacted or at risk; the prevalence, severity, economic and societal costs of these impacts; and who is most vulnerable to these impacts.
- **Pathways and mechanisms**: improving our understanding of how climate change affects mental health and, in particular, whether there are factors specific to climate change that increase mental health risks or create new experiences of mental health challenges. This includes considering biological, psychological, societal, or environmental pathways and mechanisms.
- **Mental health benefits of climate action (adaptation and mitigation)**: understanding and quantifying when and how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health.
- **Mental health interventions/solutions in the context of climate change**: identifying the most effective mental health interventions/solutions to support mental health in the context of climate change, across diverse sectors. This encompasses providing support to people already experiencing negative mental health impacts and reducing risk or severity of future negative impacts.

Table 2: Research findings relating to emerging regional needs for mental health in the context of climate change in LAC.

Research category	Overarching synthesis of findings for LAC region
Impacts, risks and vulnerable groups	**Varied risks and wide-ranging vulnerabilities:** Within LAC, risk of climate-related mental health challenges varied by geographic areas, personal demographics and sociocultural characteristics. The following groups were perceived as being most vulnerable to climate related mental wellbeing distortions and possible new onset mental health challenges: • Children • Coastal economy and farming communities • Disabled persons • Ethnic minorities • Indigenous groups • Persons with pre-existing mental health challenges • Women • Youth

	Identifying population sub-groups who experience increased vulnerability to mental health challenges caused by adverse climate events and, conversely, those experiencing resilience to these effects is equally important. Therefore, there is a need for bidirectional research exploring the dual perspectives of vulnerability and resilience. *Important refining concept:* Indigenous Peoples should be viewed as heterogenous groups. It should not be assumed that all indigenous groups have a spiritual connection. **Comments on methodology, metrics, and existing data:** Research methods must attempt to define the 'additional' mental health burden associated with climate change. Data on existing non-climate-related mental health burden is missing in some parts of LAC; therefore prevalence studies may be needed, along with surveillance, monitoring and evaluation systems to establish baselines and monitor trends.
Pathways and mechanisms	**Complex and interconnected biological and social and economic pathways:** Adverse climate events may act as primary triggers for a cascade of events leading to adverse socioeconomic outcomes which may negatively impact mental wellbeing (e.g., drought can lead to loss of food security, which can lead to forced migration and displacement, which can lead to poverty-related vulnerabilities, which can lead to distortion in mental wellbeing). It is also possible that interdependent and cascading biological, social and psychological factors secondary to exposure to adverse climate events may exacerbate existing mental health challenges. These distortions may arise due to alterations in care pathways. **Important refining concepts:** Because factors within LAC are occurring simultaneously and potentially acting synergistically, any research agenda will need to be interdisciplinary and action-oriented, simultaneously identifying and addressing the mechanisms by which the climate crisis is impacting mental wellbeing, either by creating or compounding mental health challenges.
Mental health benefits of climate action (adaptation and mitigation)	**Use of existing structures and involvement in advocacy:** There is potential benefit in using existing disaster climate response mechanisms to foster/safeguard mental health and mental wellbeing in the presence of climate threats. These response mechanisms were described as multisectoral, well-established and present at local, national and regional levels. In addition, agency and hope through advocacy was identified as a possible mental health benefit of climate action; further work is needed in this area, particularly related to youth advocacy and agency.
Mental health interventions in the context of climate change	**Collective action is necessary:** Community responses, including by religious groups and those who trust in a power beyond humanity, may be useful tools to address mental wellbeing and mental health challenges occurring secondary to climate change. Multisystem

partnering is also needed to address the combined social, economic and commercial determinants of these challenges. Dialogue participants also provided several ideas on how mental healthcare could be strengthened, including ensuring that healthcare providers offer contextualized, climate impact-relevant services.

Important refining concepts: Strengthening of mental healthcare provision and destigmatizing mental health challenges are fundamental issues within LAC. Social, behavioural and organisational research which takes into consideration climate and mental health is critical.

Research agenda

Priority research themes

Background to research categories and priority research themes

This research agenda presents an aligned vision to guide the climate and mental health field in LAC. Research priorities have been generated through extensive consultation with experts across disciplines, sectors and geographies in the region and iterated with experts regionally and globally. The priority research themes represent areas where targeted research investment could create a full picture of climate-related impacts on mental health challenges, their mechanisms, and solutions across both mental health and climate actions.

We hope that these questions will act as an inspiration to guide the direction of the research community of practice, provide investment targets for funders and focus generation of evidence to best enable policymakers and practitioners to address the climate change-related mental health needs in LAC.

Table 3: Priority research themes

Research category	Priority research theme
1. Impacts, risks and vulnerable groups	• Identifying the characteristics of people living in climate change-vulnerable populations in LAC and measuring how the mental health of these populations is affected by these vulnerabilities. • Understanding how climate-vulnerable populations in LAC perceive and respond to extreme climate events and how these perceptions and responses relate to mental health outcomes. *(This is important for populations who might not view climate change as negative, such as persons with spiritual worldviews that perceive climate change as normal or expected.)* • Investigating the association, if any, between exposure to adverse climate events and the occurrence of mental health challenges or changes in a person's baseline mental wellbeing state in LAC. • Quantifying the mental health challenge/mental wellbeing burden directly attributable to climate change among [insert population of

	interest, such as identified vulnerable groups, e.g., fishing communities] in LAC. ● Measuring the incidence/prevalence of changes in [state a specific mental health challenge, e.g., suicidal ideation] or a person's baseline mental wellbeing attributable to climate change in LAC. ● Identifying the climate events associated with the highest risk of worsening mental health challenges and/or poor mental wellbeing amongst the population in LAC. ● Identifying the sociodemographic factors that influence vulnerability to climate change-related mental health challenges in LAC. ● Identifying and understanding which lived experience factors occurring secondary to climate-related events in LAC increase the likelihood of mental health challenges. *(If explored from a qualitative lens, this question may be expanded to assess the impact of witnessing and responding to acute climate trauma, ecosystem-related loss or grief, sexual violence stemming from climate change effects, and poverty-related marginalisation/stigma/discrimination secondary to climate change, as identified in the dialogues.)* ● Identifying and evaluating the most effective ways to recruit and sample climate-vulnerable populations across large geographic areas in LAC to aid appropriate and robust data collection in climate and mental health research. ● Identifying and assessing culturally appropriate and cost-effective methods for gathering solution-oriented lived experience data about climate-related mental health in the [insert population of interest, e.g., Indigenous Peoples] affected by [insert climate event, e.g., drought] within LAC.
2. Pathways and mechanism	● Measuring the prevalence of gender-based violence occurring in climate-marginalised women and children (including persons who are homeless, migrant or housing insecure secondary to climate events) in LAC and how this affects mental health and mental wellbeing. ● Assessing the impact of heat on the bio-availability of medicines used to treat mental health conditions within LAC and the stability of these medicines during storage. *(In particular, this question explores how heat impacts the active components of medicines and how these components then interact with physiologic mechanisms.)* ● Evaluating the impact of cortisol-modifying biological mechanisms on the mental health challenges and mental wellbeing of people impacted by adverse climate events in LAC. *(Participants specifically mentioned the effect of climate-related stress on cortisol levels; effects on other stress-related hormones may also be worthy of research.)* ● Analysing the impact of [insert any climate event] on the circadian rhythm of [insert any population] in LAC and the resulting effects on mental health outcomes. *(Circadian rhythm refers to physiological sleep-wake cycles; this question*

	explores the mental health effects of disruptions that can occur in associated hormones due to changes in this rhythm from climate-related factors such as heat or displacement from housing/shelter following climate disasters, etc.)
3. Mental health benefits of climate action (adaptation and mitigation)	• Identifying the most effective interventions for fostering collective community action for climate adaptation and mitigation in ways that also improve mental health outcomes (e.g., collective action around disaster response, collective action to improve green spaces/tree planting projects, etc.) in LAC. • Identifying and evaluating the most effective interventions for promoting advocacy and agency as a tool for fostering climate resilience in young people in LAC and assessing the implications of these interventions on mental health outcomes. • Identifying the barriers to raising awareness of the impact of climate on mental health amongst climate-vulnerable populations in LAC. • Identifying and assessing culturally appropriate, accessible and affordable climate change education or awareness raising activities that could improve mental health outcomes for climate-vulnerable populations in LAC. • Identifying low-cost mechanisms for effectively sharing information regarding climate risk with vulnerable populations (including in informal and formal education settings) in LAC and evaluating how this affects mental health outcomes. *(This research should include identifying and evaluating existing information sharing structures in either mental health or climate spheres, for example leveraging disaster emergency warning systems.)* • Assessing the potential of modifying existing disaster risk communication strategies (which are used to help minimise the mental health effects of climate disasters in climate vulnerable regions) to address the mental health effects of creeping and acute climate threats across regions in LAC. • Identifying and evaluating the drought survival and/or land use management mechanisms used by Indigenous Peoples in LAC and their potential to minimise mental distress and improve mental health outcomes for populations affected by climate change. *(The climate event can be altered for this question.)*
4. Mental health interventions/solutions in the context of climate change	• Assessing the impact of daily mindfulness or spiritual practices on mental health challenges and mental wellbeing among the [insert a subpopulation, e.g., Calinago] in LAC in response to [insert climate event of interest].

- Identifying and evaluating the efficacy of medicinal plants and creole seeds used by Indigenous Peoples to minimise or reduce the mental health challenges caused by climate change in [insert specific region] and understanding the implications of a changing climate for their use and availability.

- Identifying the most effective ways for interdisciplinary and multisectoral action to support climate and mental health interventions in LAC.

- Identifying and assessing the shared (across climate and mental health) policy and funding mechanisms across the region to support research at the mental health and climate intersection in LAC.

- Understanding a) how people with mental health challenges connect to care services in LAC, b) whether this changes in the context of climate-related crises, c) if this changes, how outcomes are affected, and d) the best way to provide appropriate and continued care.

- Identifying and evaluating which collective and community-based interventions are most beneficial to protect mental health and mental wellbeing in the context of the climate crisis in LAC.

- Understanding the role of religion and spirituality (trusting in the existential) in contextualising/experiencing mental health challenges in relation to climate change in LAC, and how to appropriately account for this in intervention development and implementation.

- Identifying the most efficient multi-system partnerships to address the combined social, economic and commercial determinants of mental health challenges and/or poor mental wellbeing that may be worsening as a consequence of acute and chronic climate threats.
  (This question is included to highlight the need for multisectoral and system-level action to investigate and address the impact of the climate crisis on mental health.)

- Understanding the nature and efficacy of self-management and resilience practices of Indigenous populations with respect to [insert adverse event] and [insert mental health outcome of interest].

Action agenda

As the climate change and mental health field in LAC builds and the evidence base grows, it will be crucial to avoid perpetuating existing challenges, including: 1) disconnections across disciplines and between researchers and policy makers; 2) unequal focus on topics and geographies; and 3) siloed decision making for climate and for mental health. This action agenda sets out a shared vision as a rallying focus of the mental health and climate change field in LAC. This includes guidance on how to best support the growing community of practice, how to translate evidence to action, and the principles that should guide this approach. Enacting this agenda will require transdisciplinary effort and coordinated action across research, research funding, policy, and practice. This action agenda aims to assist this work by setting out the challenges which must be addressed, opportunities that can be harnessed, and priority actions to work towards a thriving climate and mental health field.

Regional vision for mental health in a changing climate

Vision statement

Based on the visioning exercise and conversations in Dialogues 1 and 2, participants cast a hopeful vision for the future where ***through collective corrective actions, the human impacts on climate change are reversed or minimised. Climate-related mental health outcomes are improved by reducing climate hazards and creating an enabling policy and social infrastructure, which includes participatory interdisciplinary research and better diagnostic and treatment services for climate-related mental health challenges.***

- The desired state of climate-related mental health is one in which:

    1. Policy actions to protect the mental health of the population from climate threats are implemented based on evidence produced by a regional community of interdisciplinary researchers who are working collaboratively across sectors of society (including people with lived experience);
    2. Key terms (e.g., eco-anxiety) are clearly defined in LAC context;
    3. Diagnostic criteria for climate related impacts on mental health are clearly established;
    4. Validated psychometric instruments or tools are in place for the region;
    5. Responsive pathways exist for directing people to appropriate mental healthcare in response to acute and chronic climate threats; and
    6. The provision of mental healthcare in the context of climate change occurs in tandem with mental health system reform and destigmatisation of the concept of mental health in the region.

Creating an enabling environment for research at the intersection of climate change and mental health

The identified research priorities will only be of value if they are enacted. The climate and mental health field is relatively new and rapidly growing, and now is the time to ensure that it is designed to deliver a mentally healthier future in the context of the climate crisis. In a field that spans multiple disciplines and sectors, each with different cultures and ways of working, and on a topic with low awareness in many countries, it is necessary to identify and describe what is needed to support capacity building efforts. What principles must guide the field, and what are the challenges and opportunities in the region to create an environment that would enable such research?

This section presents a synthesis of dialogue discussions on what is needed to implement the research agenda and foster an enabling research environment for climate change and mental health. This data presented with support quotes in the table below.

Table 4: Creating an enabling environment for research at the intersection of climate change and mental health

Concept	Summary	Supporting quotations from participants*
Desired State	There is a fundamental need to dissect the impact of climate change on mental health, including defining vulnerability, the concept of 'the impact' and the relationship to wider social contexts. Research should define climate-related mental health states, especially those that may have compounding factors, with further development or validation of diagnostic tools. Finally, there is a need for research to inform the organization and delivery of sustainable mental health services related to or triggered by the climate crisis. This will require community-based participatory approaches to research.	*"So, I see the research as generating a better understanding of the sort of spatial dynamics of mental health in a climate change context. In other words, I think we are assuming that there is some difference in the environment of mental health during climate crises in other spaces. That seems to be an assumption."* – **English breakout 1, Dialogue 2** *"Mental health is pretty dirty, actually. It's very hard for you to clean. For example, just eco-anxiety… I think that the creation and validation of instruments is one thing, I think that institutional diagnosis... And the hardest thing I think is diagnosis of eco-anxiety which I know has scale, but it's also not validated. You have to validate. It's very early that."* –**Portugese breakout, Dialogue 2** *"We need the academic researchers, the epidemiologists, the social scientists to look at the social impact as well [unclear] affiliated with the clinical or social determinants of health."* – **English breakout 2, Dialogue 2**
Challenges	• Lack of consensus on key terms • Lack of validation of tools	:*"I think one of the things we're not really picking up on is things like post-traumatic*

	• Absence, or limited utility, of surveillance systems or registries for mental health data • Possible lack of understanding of the presentation of delayed mental health manifestations post-disaster, leading to late diagnoses • Funding	*stress disorder, which happens long after.”* – **English breakout room 2, Dialogue 2** *“How do the symptoms evolve? How do they present at the time of disaster, after a period of time and after a longer time?”*– **Post-dialogue survey, Brazil region** *“I know that suicide data is hard for you to measure. And if you're trying to make mental health impacts or... We don't have any very good network and mental healthcare. So, how do you measure and be able to make this overlap”* – **Breakout, Dialogue 2** “*Funding Agencies could turn their eyes to this, because everything that is said about climate change is very long-term things, but very much focused on environmental issues of rescuing biodiversity, rereading food production processes... food and nutrition sovereignty and security, which directly impacts, in my view, mental health”* – **Breakout, Dialogue 2** *“Train healthcare providers to notify the International Classification of Disease (ICD-11) linked to climate change. For instance, insomnia due to heat wave: notify both ICD G47”* – **Post-dialogue survey, Brazil region**
Opportunities and enablers	• **Partnering with existing institutions**, such as Universidade de Sao Paulo or Federal University of Rio Grande do Sul, to design and execute research in academic and nonacademic settings **Leveraging currently commissioned** work by large international groups in related areas, like food security, in the region	*“We have the students, we have the universities, in this case, we are all from academia, we have a health system. We already have the structure. I think that's an opportunity, it's an enabler”* – **Portuguese breakout, Dialogue 2** *“I really wanted to use the students, because the students, when we taught the class, they came with a lot of eco-anxiety, with a lot... One girl even said, "I want to drop out of medical school because I'm going to make people live longer and the planet is going to suffer. So, I think I'm going to quit medicine." They're talking about it a lot. I was thinking more about researching them or nearby communities that are suffering, such as Rio Grande do Sul.”* – **Portuguese breakout, Dialogue 2**

Relevant potential partners	Key stakeholders involved in multi-sectoral research should include: • Civil society organisations • Lived experience groups • Varied health professionals • Youth	*"For that, obviously we would have to have the academic and research partners and stakeholders. Those don't have to necessarily be housed in a university. We need to also think about other research entities that conduct research in this area."* – **English breakout room 2, Dialogue 2** *"........and we're talking about having the community members, including the youth, as important stakeholders or partners, more partners than stakeholders, in the research."* – **English breakout room 1, Dialogue 2** *"It's a multidisciplinary thing, I think. Professors from other institutions, because, in fact, there are people who are more prepared in other places, more mobilized than us"* – **Portuguese breakout room 2, Dialogue 2**

**Some quotes cover dual concepts.*

Priority next steps/recommendations

Given the multiple factors influencing the interaction between climate change and mental health in LAC, funders, researchers and policy makers should urgently seek to adopt a multisectoral approach which fundamentally views stewardship of the earth's resources as being inseparable from continued human development and wellbeing.

Funding institutions should:

- Create shared pools across institutions from diverse sectors to increase available funding/resources (e.g., pooling resources from traditional funders of health and traditional funders of security response research).
- Create dedicated allocations for climate and mental health within existing explicitly long-term funding streams (e.g., food policy research).

Funding and research institutions should prioritise projects that:

- Create or augment mental health and climate surveillance systems; this will aid in understanding the baseline epidemiological climate and mental health interactions in the region.
- Create and validate climate and mental health diagnostic tools (e.g., those needed for eco-anxiety).
- Contain practical components for increasing mental health and/or mental wellbeing in the context of climate change at the community level (preferably with demonstrable sustainability and high community impact).

Translating a growing evidence base into action that can respond to the mental health impacts of climate change

The current evidence base on the interconnections between climate change and mental health compels action in policy and practice to protect escalating mental health needs and promote co-beneficial climate actions. How can current evidence and new insights created through implementing this agenda best translate into changes to policy decisions and practices across both climate and mental health regional spaces?

This section presents a synthesis of dialogue discussions and survey results, setting out the challenges and opportunities to translate evidence generated through research into policy and practice.

Table 5: Translating a growing evidence base into action that can respond to the mental health impacts of climate change*

Concept	Summary	Supporting quotations from participants**
Desired state	The body of evidence would be used to influence or support decision makers across the multiple policy spheres that impact the nexus between climate and mental health. This will require education, engagement, communication and involvement with policymakers and various actors across society (e.g., media to attract buy-in, funding and general support to raise community awareness).	*"Advocacy with the authorities proposing CC measures for vulnerable communities in the Caribbean and Hati in Particular (Public Health-Mental Health unit and the Ministry of Environment). These issues are not currently on the agenda… research results are presented as well as the effects on communities affected by disasters linked to climate change including the impact on displacement (internal migrations)."* – **French breakout room, Dialogue 2**
Challenges	• Relative **newness of mental health and climate change** as a public health issue of significance; therefore, the issue may be low on political and other social action agendas • General **societal stigma around mental health**, which presents challenges to awareness and priority setting at the policy level	*"And related to climate change, it's hard to make that connection. Why has climate change impacted my quality of life or my mental health? In quality of life, it's easy for you to sometimes make an association. In food, it's easy to make an association now. In mental health, it's hard because it feels like it's kind of a veiled thing.....First, mental health is a veiled thing, I think, in our population. At least,*

		Brazilian. I see it as something like that. Addressing the topic of mental health is something very serious. You can't keep saying it anywhere, anytime, anytime. And even associating it with the issue of climate change is even more complex….People are always saying, 'We need to talk about it.' Until recently, we couldn't even talk about suicide, for example. We couldn't even talk about depression." – **Portuguese breakout room, Dialogue 2**
Opportunities and enablers	• **Harnessing current interest in the field** • **Highlighting acute crisis climate situations** to highlight the urgency of a response and the need for long-term solutions • **Leveraging community preparedness and collective action**	*"Time is a catalyst and a stress- for example- rising sea levels, people are aware that the loss of territory is coming. Impact on seasons and rainfall, harvests, mangroves that protect coastal towns, are also impacted; these changes can be catalyst for decision-makers to take action."* – **French breakout room, Dialogue 2**
Relevant potential partners	• **State/government stakeholders,** including ministries of the environment and health services • **Non-governmental stakeholders,** including representatives from schools, universities, coastal and mountain communities, people with disabilities, mental health organisations, churches, farming communities and fisher peoples, women and victim's associations, and grassroots organisations • **Private sector** businesses with the potential, or perceived potential, to negatively impact ecosystems in the region (e.g., fertiliser producers, agribusiness and large consumers of land and water) • **International entities,** such as UN agencies, women's organisations, medical associations, psychological associations and research centres	*"All levels of government, ranging from the mayor's office to the president….These would be facilitating elements… facilitators in the creation of public policies or in the implementation or creation and implementation of public policies?"* – **Portuguese breakout room, Dialogue 1** *"…you want to bring in stakeholders who will be influential in policy development. So, it's not like you do your research, you have your findings, you have a policy brief, and now you go to the people with the*

		power. You want them on board from the beginning, so that they can buy into your project, and they become a part of the solution from day one." – **English breakout room 2, Dialogue 2** *"You can create these policies, but you still need the support of the community to really help drive it into action"* – **English breakout room 2, Dialogue 2** *"To branch off with the social media, who our partners can be, I guess that goes hand in hand with the social media influencers to really get the attention of the youth because they play a very major role in the development of our future and our present."* – **English breakout room, Dialogue 2**

**Please note that there is overlap in findings, which reflects the likely synergystic approach that will be needed to operationalize the research agenda.*

***Some quotes cover dual concepts.*

Priority next steps/recommendations to investors and actors

Policy makers should:

- Create mechanisms for enhancing medical education and training in the climate and mental health field.
- Incorporate climate and mental health into priority health system policies/components (e.g., climate and mental health as part of universal health coverage, primary healthcare and pandemic responses).

Funders, policy makers and researchers should:

- Reflect on and design culturally appropriate ways of translating evidence across varied geographic settings, population groups and languages.
- Engage with media and other actors to support the translation of evidence.
- Invest in organised advocacy campaigns on climate and mental health in the region to reduce stigma.

Discussion: strengths, limitations and next steps for the research and action agenda

The CCM project was a novel exercise and a tremendous opportunity to gather lived experience knowledge across a large, multi-lingual, multicultural region with countries at all levels of human development. Strengths and limitations of the project are briefly outlined below, with a view to possible next steps.

Strengths

- Surfacing needs and experiences not previously heard or captured in literature.
- Diversity of experts brought together and sharing within and across regions.
- Opportunity for multilingual academic and lived experience exchange, which may have been limited by language barriers that have traditionally created silos in the region.
- Meaningful engagement of lived experience expertise guiding this work.
- Raised awareness of the nexus of climate and mental health in LAC.
- Personal value of being involved for participants (including safe spaces to share what they and their communities face as well as building their skills, capacities and connections).
- Design of a new methodology for cross-cultural lived experience research and management of large qualitative datasets in LAC.

Limitations

- Balancing the urgency to develop a research and action agenda for this rapidly growing field with meaningful convening, relationship building and moving at the pace of trust, especially with groups frequently subjected to extractive research processes (i.e., Indigenous Peoples and Nations).
- Addressing the magnitude and embeddedness of the challenges this work seeks to start to overcome (disconnected, uneven, siloed sectors and research processes).
- Challenges of virtual convening (e.g., technology a barrier for some participants).
- Building processes (e.g., data sharing and ethics) from scratch (also a strength).
- Challenges of balancing need for standardisation with local contextualisation and need to ensure use of local terminologies and cultural understandings (this could also be a strength regarding how we developed methods globally with working groups having opportunities for representation across all regions).
- Stigma/taboo of mental health and issues of gender identity within the region. It is possible that survey design elements linking name, gender and mental health status (although being confidential) might have impacted response rates.

Conclusion

Participants from LAC perceive and experience multiple varied threats to mental health and mental wellbeing by creeping and acute climate events. Given the increasing intensity of these threats, we need to collaboratively establish fundamental principles for designing evidence-based climate and mental health interventions, supported by a platform for dissemination and implementation of these interventions. Achieving these goals will require a systems-thinking approach, operating across sectors and disciplines and should include a wide range of stakeholders from the formative stages. The task will require significant support from regional and international actors and funders but can be achieved through unified action and respectful and inclusive ways of working.

Hearing from the Regional Community of Practice

At the end of the dialogue process we asked participants to provide thoughts on what they hoped would emerge from the RCoP. In honour of their thoughts, we provide here a list of all the comments as a virtual wall of memorium of the hopes of these pioneers regarding mental health and climate in LAC.

- Yo haré procesos de incidencia en la organización donde trabajo para que se aborden tema de salud mental y cambio climático para el beneficio de las comunidades indígenas (I will carry out advocacy processes in the organization where I work so that mental health and climate change issues are addressed for the benefit of Indigenous communities).
- I will share a policy brief on climate change and mental health on children.
- I will contribute by advocating globally for policies that empower youth to be educated trained and employed to solve the issues.
- Yo contribuiré a divulgar lo aprendido ydiscutido. (I will contribute to disseminating what has been learned and discussed.)
- I will maintain the connections I made today in order to take action on issues related to mental health and climate change effects.
- I will keep highlighting the voices of youth in the climate and mental health agenda. Also, I will commit to start raising awareness about the connection of this matter in mi city and the spaces I am enroll.
- I'll be thinking about how my contribution can be more effective by working with different players to implement better solutions.
- I will make impact on my community.
- I will seek to integrate the mental health related issues and CC integrated in the Caribbean Community Risk Information Tool (CCRIT) and other related community resilience worl.
- I will ensure that contexts with the least amount of capacity are considered.
- I will follow up with the aftermath of this project and assist with anything that I can along the way.
- Yo hare mis próximas investigaciones con enfoque de salud mental. (I will do my next research with a mental health focus).
- I will continue considering mental health as a key element to sanity.
- De esta comunidad me gustaria ver acciones concretas en el campo de salud mental y cambio climático. (From this community I would like to see concrete actions in the field of mental health and climate change).
- Out of this community, I would like to see practical ways forward for relevant research and policy development.
- Out of this community I would like to see lasting partnerships.
- Out of this community I would like to see real policy change across the region.
- I would like to see the dashboard/platform be maintained after this research cycle ends.
- More scientific evidence generated, and more policies implemented.
- De esta comunidad quiero contar con apoyo técnico, político y si fuera posible económico para algunas acciones. (From this community I want to have technical, political and, if possible, economic support for some actions).

- I would like to see a clear systematization of the main findings and how we can start actions.
- Out of this community I would like to have more outreach work dealing with mental health and climate changes.
- Out of this community I would like to see opportunities for collaborative research.
- I would like to see strategic efforts to have Mental health issues integrated into existing development and social policies.
- Out of this community I would like to see the creation of avenues for change available to youth, bridging of gaps to organizations that already exist and mental health outreaches to persons in rural areas.
- Out of this community I would like to see our own community growing in awareness and hope.

Glossary of terms

For a glossary describing relevant concepts and key words for the Connecting Climate Minds research and action agendas, please download from here.

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This is the independent work of the authors with the support and input of the Connecting Climate Minds team.

Acknowledgements

The Latin America and the Caribbean Connecting Climate Minds team thanks everyone who contributed to the production of this research. The varied geography within the region made it challenging for some participants, and we do express our sincerest thanks to those who gave of their time and expertise.

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Further the work would not have been accomplished without the expertise, diligence, interest and support of the Co-Convenors, Lived Experience Persons and Youth Ambassadors.

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The team further thanks Dr. Natalie Greaves for her technical expertise and leading the analysis of the qualitative data presented in this report.

Funding

This project was funded by Wellcome.

Conflicts of interest

The authors have no conflicts of interest to declare.

Appendix

Appendix 1: Recommended points for policy intervention based on the scoping review of literature

1. Improve organization and delivery of mental health services including those that address the needs of Indigenous populations mental health infrastructure generally.[8]

2. Develop and implement programmes for building capacity among community members to deliver Psychological First Aid.

3. Develop and deliver training for health professionals on the links between mental health and climate events.[24]

4. Establish continuity of care mechanisms for people with mental health during adverse weather events.[24]

5. Develop and implement programmes to facilitate mental wellness through anticipatory climate change adaptations.[24]

6. Implement passive surveillance mechanisms for mental health including the International Classification of Disease (ICD-10 or 11) diagnostic tool.[10]

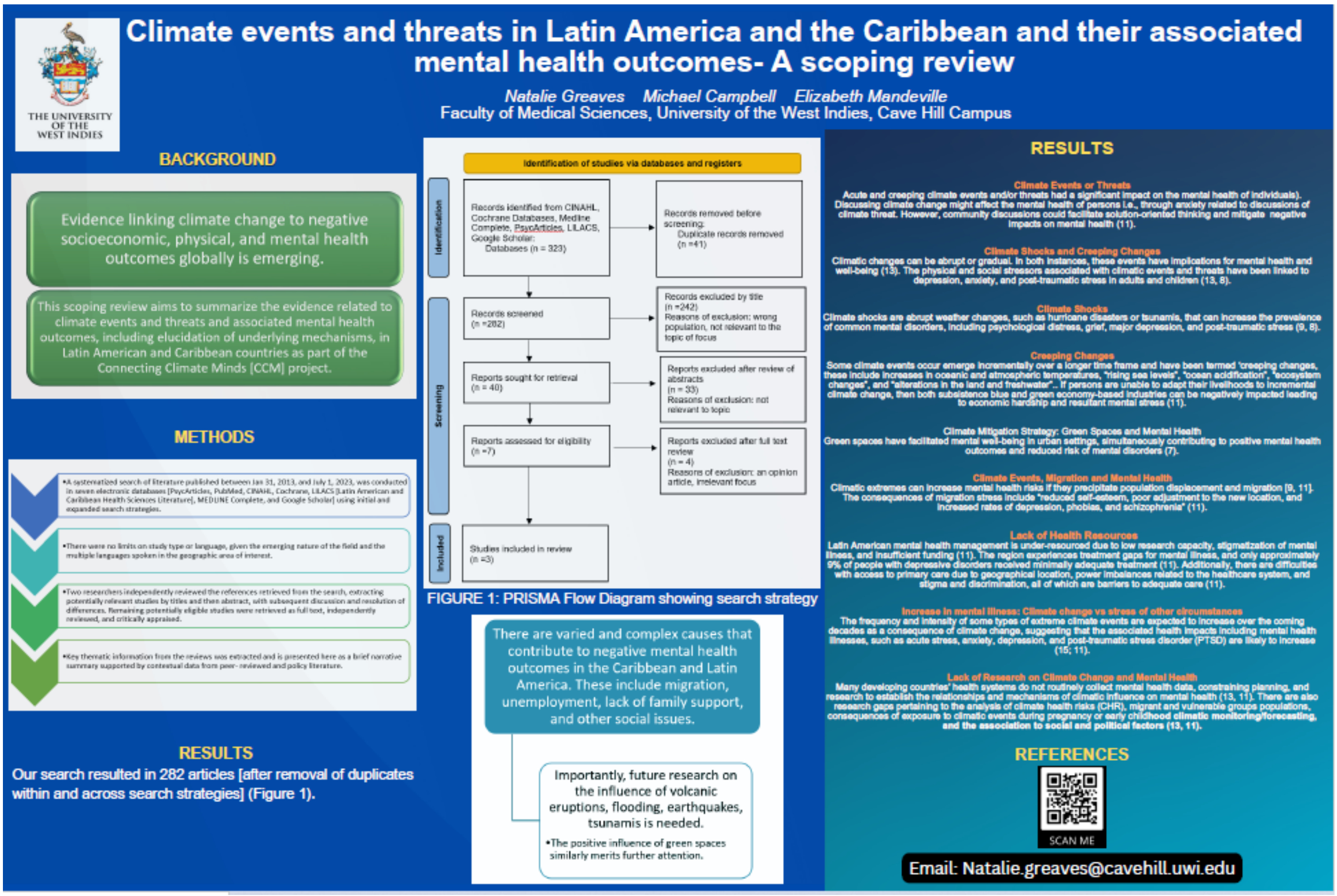
7. Develop community-facing health promotion programmes on the interaction of mental health and climate change.[5]

8. Discuss climate change to open conversations about the detrimental mental health impacts.[9]

9. Use a participatory process for the development of public policy on climate change and resilience, recognising that people have a right to be involved in local-to-global planning, decision making and action.[5]

Appendix 2: Poster summarising elements of the results of the narrative scoping review

- Summary of methodology- The review examined literature published between Jan 31, 2013 and July 1, 2023, and was conducted in seven [7] electronic databases (PsycArticles, PubMed, CINAHL, Cochrane, LILACS [Latin American and Caribbean Health Sciences Literature], MEDLINE Complete, and Google Scholar[1]) using initial and expanded search strategies. There were no limits on study type or language, given the emerging nature of the field and the multiple languages spoken in the geographic area of interest.



Climate events and threats in Latin America and the Caribbean and their associated mental health outcomes- A scoping review
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BACKGROUND
Evidence linking climate change to negative socioeconomic, physical, and mental health outcomes globally is emerging.
This scoping review aims to summarize the evidence related to climate events and threats and associated mental health outcomes, including elucidation of underlying mechanisms, in Latin American and Caribbean countries as part of the Connecting Climate Minds [CCM] project.
METHODS
•A systematized search of literature published between Jan 31, 2013, and July 1, 2023, was conducted in seven electronic databases [PsycArticles, PubMed, CINAHL, Cochrane, LILACS [Latin American and Caribbean Health Sciences Literature], MEDLINE Complete, and Google Scholar] using initial and expanded search strategies.
•There were no limits on study type or language, given the emerging nature of the field and the multiple languages spoken in the geographic area of interest.
•Two researchers independently reviewed the references retrieved from the search, extracting potentially relevant studies by titles and then abstract, with subsequent discussion and resolution of differences. Remaining potentially eligible studies were retrieved as full text, independently reviewed, and critically appraised.
•Key thematic information from the reviews was extracted and is presented here as a brief narrative summary supported by contextual data from peer- reviewed and policy literature.
RESULTS
Our search resulted in 282 articles [after removal of duplicates within and across search strategies] (Figure 1).
Identification of studies via databases and registers
Identification
Records identified from CINAHL, Cochrane Databases, Medline Complete, PsycArticles, LILACS, Google Scholar: Databases (n = 323)
Records removed before screening: Duplicate records removed (n =41)
Screening
Records screened (n =282)
Records excluded by title (n =242) Reasons of exclusion: wrong population, not relevant to the topic of focus
Reports sought for retrieval (n = 40)
Reports excluded after review of abstracts (n = 33) Reasons of exclusion: not relevant to topic
Reports assessed for eligibility (n =7)
Reports excluded after full text review (n = 4) Reasons of exclusion: an opinion article, irrelevant focus
Included
Studies included in review (n =3)
FIGURE 1: PRISMA Flow Diagram showing search strategy
There are varied and complex causes that contribute to negative mental health outcomes in the Caribbean and Latin America. These include migration, unemployment, lack of family support, and other social issues.
Importantly, future research on the influence of volcanic eruptions, flooding, earthquakes, tsunamis is needed.
•The positive influence of green spaces similarly merits further attention.
RESULTS
Climate Events or Threats
Acute and creeping climate events and/or threats had a significant impact on the mental health of individuals). Discussing climate change might affect the mental health of persons i.e., through anxiety related to discussions of climate threat. However, community discussions could facilitate solution-oriented thinking and mitigate negative impacts on mental health (11).
Climate Shocks and Creeping Changes
Climatic changes can be abrupt or gradual. In both instances, these events have implications for mental health and well-being (13). The physical and social stressors associated with climatic events and threats have been linked to depression, anxiety, and post-traumatic stress in adults and children (13, 8).
Climate Shocks
Climate shocks are abrupt weather changes, such as hurricane disasters or tsunamis, that can increase the prevalence of common mental disorders, including psychological distress, grief, major depression, and post-traumatic stress (9, 8).
Creeping Changes
Some climate events occur emerge incrementally over a longer time frame and have been termed 'creeping changes, these include increases in oceanic and atmospheric temperatures, "rising sea levels", "ocean acidification", "ecosystem changes", and "alterations in the land and freshwater".. If persons are unable to adapt their livelihoods to incremental climate change, then both subsistence blue and green economy-based industries can be negatively impacted leading to economic hardship and resultant mental stress (11).
Climate Mitigation Strategy: Green Spaces and Mental Health
Green spaces have facilitated mental well-being in urban settings, simultaneously contributing to positive mental health outcomes and reduced risk of mental disorders (7).
Climate Events, Migration and Mental Health
Climatic extremes can increase mental health risks if they precipitate population displacement and migration [9, 11]. The consequences of migration stress include "reduced self-esteem, poor adjustment to the new location, and increased rates of depression, phobias, and schizophrenia" (11).
Lack of Health Resources
Latin American mental health management is under-resourced due to low research capacity, stigmatization of mental illness, and insufficient funding (11). The region experiences treatment gaps for mental illness, and only approximately 9% of people with depressive disorders received minimally adequate treatment (11). Additionally, there are difficulties with access to primary care due to geographical location, power imbalances related to the healthcare system, and stigma and discrimination, all of which are barriers to adequate care (11).
Increase in mental illness: Climate change vs stress of other circumstances
The frequency and intensity of some types of extreme climate events are expected to increase over the coming decades as a consequence of climate change, suggesting that the associated health impacts including mental health illnesses, such as acute stress, anxiety, depression, and post-traumatic stress disorder (PTSD) are likely to increase (15; 11).
Lack of Research on Climate Change and Mental Health
Many developing countries' health systems do not routinely collect mental health data, constraining planning, and research to establish the relationships and mechanisms of climatic influence on mental health (13, 11). There are also research gaps pertaining to the analysis of climate health risks (CHR), migrant and vulnerable groups populations, consequences of exposure to climatic events during pregnancy or early childhood climatic monitoring/forecasting, and the association to social and political factors (13, 11).
REFERENCES
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Appendix 3: Dialogue agendas

Dialogue 1

Time (mins	What	How
0-20	**Welcome and introductions**	
20-35	**Visioning Exercise**	Visioning Exercise: "Headlines" on Climate Centre Good Games platform Participants share what they are hoping to get from being part of Connecting Climate Minds, and why they are invested in this topic

		Imagine we are in 2030 and the CCM project has received an important prize for its achievements in bringing CC/MH together in your region and making change for climate and mental health together through research and action. You are interviewed by a journalist as you are celebrating winning the prize Create / vote on and review Headlines created on the Good Games Platform
35-50	**Scene setting**	Brief context to the links between climate change and mental health What are we hearing in the region that we can come together and respond to: sharing highlights of pre-scoping Presentation on Connecting Climate Minds and Q&A
50-95	**Breakout session 1: Regional needs**	Host sets scene for breakout discussion Share climate hazards overview for the region **Breakout groups** Facilitator to lead the group in a discussion on the mental health challenges arising from climate change in the region. Questions posed to participants were: ● *From your perspective, what are the climate hazards that concern you the most in terms of mental health and wellbeing?* ● *For each hazard, discuss the following:* ○ *Who is being impacted? Are there any groups more at risk?* ○ *What are the mental health and wellbeing outcomes associated with this hazard?* ○ *How does this hazard specifically affect mental health and wellbeing; what are the mechanism e.g. housing, livelihoods etc* ○ *Encourage people to look at different types of mechanisms e.g. social, biological, cultural, environmental* ○ *Climate adaptation/mitigation actions - what is being done, what can be done?* ○ *Actions that are being/could be taken to improve or promote mental health and wellbeing in this context* Brief plenary highlights
95-105	**Break**	
105-110	**Sneak preview**	The cartoon artist shares one first draft reflecting some of the content of the session so far. Host invites participants to share in the chat how the image resonates.
110-160	**Breakout session 2:**	**Breakout groups** [same participants, facilitator and note-taker as first breakout session]

	Evidence gaps and research **priorities**	Looking back at the hazard table from Breakout session 1 as a starting point, participants are guided to start thinking about **where more research is needed**. Hosts invite 2 or 3 facilitators to share some brief reflections
160-170	**Cartoon reflection on dialogue output**	Participants individually review each cartoon, and share their reflections. This can range from: suggestions for changes to make it more accurate, to sharing how it relates to our reality, or any other insights.
170-180	**Wrap-up and next steps**	Host shares a high level recap of the session, thanks everyone and celebrates this first coming together of the regional community. Participants complete brief survey to provide feedback on the dialogue

Dialogue 2

Time (min	**What**	**How**
0-5	**World Map Exercise**	Host asks participants to place a "star" icon where they currently are and then take the "heart" icon and place it where they would like to visit one day or a place they have been and really loved/feel especially connected to.
5-15	**Welcome and introductions**	
15-30	**Sharing updates**	Presentation on global and regional progress and outcomes of dialogue 1, including vision, priority research themes.
30-50	**Priority Research Theme Feedback**	**Jamboard shared on screen.** Host introduces priority research theme activity and asks people to consider the following questions: ***Research questions*** ● *Is this a valuable theme? Would you add or change anything?* ● *Share any ideas on datasets, metrics or methods that could be applied to help address this.* ***Research topics*** ● *What would be most helpful to know about this topic to understand and respond to it?* ● *What would you want to know first?* Music plays and participants work independently on Jamboard at their own speed.

50-55	**Scene setting for breakouts**	Host sets the scene for the breakout discussion. Interested in the 'how', which includes the principles and desired outcomes for research, policy and practice in the region. The discussion will enable targeting of action and investment to both implement the research agenda in the region, and to ensure evidence on the climate and mental health nexus is translated into policy and practice
55- 90	**Break out 1**	Eight breakout groups. The first 4 on the theme ***“Creating knowledge through research”*** (e.g. How can we best implement the research agenda?) and the other 4 on ***“Fostering evidence-based policy and action”*** *(e.g. How can we translate the growing evidence base to action in policy and practice?).* Each group uses a Jamboard to discuss: 1) What does this look like when it is done well? 2) What opportunities & enablers exist? 3) What challenges must we overcome? 4) Who should our partners and stakeholders be? Brief plenary highlights
90-100	**Break**	
100-105	**Draw your feelings**	Host welcomes everyone back and invites all who’d like to join to grab a piece of paper and a pen/marker (or digital) - and draw how they are feeling today. Anyone willing is invited to show their image - share some reflections.
105-125	**Spectrum Mapping**	Host introduces and explains the activity. **Share screen of Jamboard.** Designed to reveal the diversity of perspectives and options around different concepts and to organize them into a meaningful spectrum. This activity aims to illuminate the group’s range of perspectives and to organize those perspectives into a continuum. **Question:** *What do these concepts mean to you?* Focusing on one board at a time, host asks everyone to silently generate a point-of-view around that concept and write it on a sticky note on the Jamboard. People can add more than one. Once the sticky notes are posted, the host works with the group to sort them into a horizontal range of ideas. Sticky notes that express similar perspectives or options should go next to each other. Sticky notes that seem to be outliers should stand alone; they may sometimes end up defining the limits of the range. Repeat with the remaining concepts. At the end, ask for observations and insights on the lay of the land. And if any perspective or option has been excluded. If so, add it and re-sort as necessary.

125 - 155	**Break out 2**	Host introduces the second round of breakout rooms and provides a 2-minute summary of the Jamboard contributions from the first breakout. Original 8 groups reconvene and expand and add to their initial thoughts. Groups are allowed to continue on the theme they were already discussing or switch to the other theme.
155- 175	**I will ... I want...**	Host asks everyone to personally reflect about how they will contribute to this space, and share an "I will" statement in the chat, but only press enter when told to do so. After a minute, host invites everyone to press enter at the same time to make statements appear. This is repeated with a statement about "Out of this community I would like to see...".
175	**Wrap up and next steps**	Host shares a high level recap of the session, thanks everyone and outlines next steps. Participants complete brief survey to provide feedback on the dialogue

Appendix 4: Analysis methodology for Dialogues 1 and 2

Analytic Data reduction steps-Dialogue 1

- Four analysts (NG, SM, SJ, and SS) from varied health sciences backgrounds and who were involved in either the design and or execution of the dialogue coded the data and formulated the thematic concepts.
- The analysts first established consistency in interpretation of codes and coding method by independently coding 4 pages of dialogue with subsequent discussion and resolution of differences. Once this quasi- interrater reliability was established the transcript data set was divided amongst the analysts. Each transcript was then coded by one analyst with subsequent quality check for consistency in application of the coding frame by a second analyst.
- A group discussion was held after the coding of 4 FG transcripts to discuss emerging themes and the steps to be taken in condensing the data using a framework/ matrix analysis method (in this instance the goal was to progress from basic to organizing themes, to an overarching global theme).

|

Analytic Data reduction steps-Dialogue 2- Jamboards

Google Jamboard sheets from the refinement of research question activity were divided among the analysts (NG, GB, SM, SS and SJ). The analysts then compared all responses to a given question with original thematic summaries from dialogue 1, with the aim of identifying any critiques, changes (including expansions) to the questions/ thematic scope of the questions as articulated). The changes as articulated by participants were then summarized as eight priority research themes reflected in 26 questions to inform the research agenda in the LAC. (These questions were shared worth participants in the post dialogue survey).

Appendix 5: Criteria for selection of priority research themes

Priority research theme selection criteria was generated based on discussion with experts globally across sectors, disciplines, and by drawing on selection criteria used in other research priority setting exercises.[25] Criteria was applied by consultation within the analysis team.

- Does the research theme answer an existing evidence gap?
- Does the research theme have potential to inform decision making in policy and practice?
- Does it reflect one or more of the following:
    - identified greatest mental health challenges linked to climate change in the region
    - identified greatest climate hazards in the region
    - evidence policymakers most need to address emerging needs / overcome barriers to action
    - alignment with what the experts by lived experience voiced as challenges in the region
- Could it be feasible to undertake this research?
- Is it framed to be understandable, specific and practical for the research community to meaningfully address? Could a researcher or a funder with relevant expertise understand this theme?
- Is it specific enough to be meaningful and guide focussed research (if too broad or vague it won't help target resources meaningfully)
- It helps funders invest in this area (consider what is regionally relevant here)

Appendix 6: Coding frameworks for the research agenda and action agenda

Research Category	Sub-categories

1. Impacts, Risks and Vulnerable Groups This category is about improving our understanding of the ways in which mental health is affected by climate change. For example, what mental health outcomes are impacted or at risk, the prevalence, severity, economic and societal cost of these impacts, and who is more vulnerable to these impacts. This category also includes the ways we can go about getting this improved understanding of mental health impacts of climate change - the methods and metrics we need to assess and monitor mental health in ways that are relevant to climate change, contextually appropriate, comparable etc.	*Cross cutting considerations to keep in mind for all sub-categories:* *Timeframe* *Geographical variation*
	1.1. Research that focuses on the **prevalence, severity and nature** of the experience of different mental health outcomes/challenges/experiences affected by different aspects of climate change. This may include research to understand the emergence of climate-specific mental health experiences and their relationship to already defined mental health challenges.
	1.2. **Quantifying the fraction of mental health burden** (including mortality) caused by climate change.
	1.3. Understanding the **risk factors to mental health** that are caused or affected by climate change as well as **protective factors**.
	1.4. Identifying **population sub-groups** (e.g. demographics, livelihoods, life stage, pre-existing mental health challenges) who experience **increased vulnerability** to mental health challenges caused by climate change, and conversely those experiencing **resilience** to these effects (i.e. vulnerable groups).
	1.5. **Quantifying the cost** (e.g. economic, social) of the additional mental health burden caused by climate change and insufficient climate action.
	1.6. Methods research to identify the **most appropriate ways to assess and monitor the mental health impacts of climate change** [including adapting pre-existing scales, creating new ones, determining appropriate mental health metrics and indicators for inclusion in global processes like Lancet Countdown]. This can also include the need for cross-cultural validation and development of culturally appropriate metrics.

2. Pathways and mechanisms This category is about improving our understanding of *how* mental health is affected by climate change. We are interested in research themes that can help identify, categorise and understand the range of ways that climate change or climate action may act to affect mental health. This can include considering pathways and mechanisms that are biological, psychological, societal or environmental in nature, and may be direct or indirect. Note that mechanisms can include mechanisms to the development, maintenance, and/or resolution of mental challenges, so this includes also mechanisms relevant to guide development or understand workings of interventions	*<u>Cross cutting considerations to keep in mind for all sub-categories:</u>* *What factors are linked with increased vulnerability or increased resilience for the associated mental health outcomes.*
	2.1. Categorising and understanding the **societal mechanisms** by which climate change negatively impacts mental health [e.g. changes to livelihoods, disruption to cultural practices, food and water insecurity, forced migration, political factors]
	2.2. Categorising and understanding the **environmental mechanisms** by which climate change negatively impacts mental health [e.g. air pollution, reduced exposure to biodiversity]
	2.3. Categorising and understanding the **psychological mechanisms** by which climate change negatively impacts mental health [e.g. how temperature affects cognitive changes relevant to symptoms of mental health challenges].
	2.4. Categorising and understanding the **biological mechanisms** by which climate change negatively impacts mental health [e.g. impacts of psychotropic medication on thermoregulation, neurodevelopmental factors].
	2.5. Understanding mechanisms whereby climate action or mental health **interventions benefit climate and mental health** (i.e. co-beneficial mechanism).
	2.6. Methods research to identify the **most appropriate ways to assess and monitor pathways and mechanisms** by which climate change negatively impacts mental health and wellbeing (e.g. systems mapping across disciplines)
3. Mental health benefits of climate action [adaptation and mitigation] This category is about how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health. This includes quantifying costs and benefits of climate action for mental health, understanding what is needed to support better alignment between climate action and mental health	3.1. **Identifying climate actions** that integrate or align with mental health benefits [co-beneficial climate actions, e.g. increased tree cover in cities]
	3.2. **Quantifying co-benefits of climate action** for mental health (including number of people experiencing the benefit, size of effect, economic considerations).
	3.3. Exploring how the mental health costs and benefits of climate action may **differ across population sub-groups** (e.g. demographics, livelihoods, life stage)
	3.4. Understanding the **governance structures/decision support tools** that enables alignment of action for climate change and for mental health across sectors
	3.5. Mapping and monitoring the integration of mental health within **adaptation and mitigation policies** across sectors [e.g. National Adaptation

action, and identifying where this integration is already happening across strategies and policies.	Plans, energy, transport, food, water, agriculture]
	3.6. Exploring opportunities for mental health to be integrated into **other climate priority areas** i.e. loss and damage and climate finance.
	3.7. Determining best approaches for climate action (e.g. emissions reductions or climate adaptation) **within the mental health sector** (ensuring psychiatric facilities can be kept cool in heat waves; green space projects in mental healthcare facilities)
	3.8. Methods research to identify the **most appropriate ways to assess and monitor mental health benefits of climate action** [e.g. place-based approaches, methods for attributing and quantifying co-benefits, methods for assessment of the mental health implications of decisions in other sectors]
4. Mental health interventions/solutions in the context of climate change This category is about identifying the most effective mental health interventions/solutions to support mental health in the context of climate change. This might be about providing support to people already experiencing negative mental health impacts, or about reducing risk or severity of future negative mental health impacts. This includes learning from knowledge held by different disciplines, communities and cultures, understanding how existing mental health interventions are affected by climate change, identifying and evaluating existing interventions that are relevant to the context of climate change, and developing new interventions. Interventions are relevant	<u>*Cross cutting considerations to keep in mind for all sub-categories:*</u> *LEVEL (e.g.)* *Individual, Family, Community, Systems* *MECHANISM (e.g.)* *Biological, Psychological, Social, Environmental* *SECTOR (e.g.)* *Education, Healthcare, Public Health* *Effectiveness considerations include impacts across different population groups, and implementation considerations might include providers, cost and time.*
	4.1. Understanding different **ways of knowing, being and doing** in different cultures and communities that can build individual, community and ecological resilience
	4.2. Understanding how **existing mental health interventions are affected** by climate change
	4.3. Identifying and evaluating mental health interventions that are **already designed for or relevant to the context of climate change** and/or integrate climate change considerations

at all levels (individual, family, community, systems) and across sectors.	
	4.4 Amending, implementing and evaluating relevant **mental health interventions from other settings** to be appropriate for climate-related impacts?
	4.5. Co-designing, implementing and evaluating **novel interventions** that meet climate-related mental health needs
	4.6. **Comparing** cost-effectiveness, implementation considerations, and effectiveness across interventions for a particular setting and particular population group to determine "best buys"
	4.7. Identifying, developing and evaluating approaches to **awareness-raising and capacity building** to upskill workforces to recognise and respond to the mental health impacts of the climate crisis (e.g. mental health professionals, emergency responders)

Appendix 7: Data collection and storage

Dialogues

Dialogues were conducted virtually on Zoom. Dialogues and breakout groups were recorded and transcribed by third party providers (Way with Words and Absolute Translations) and Zoom chat comments were saved. In Dialogue 1, Word documents were used to capture notes from breakout discussions. In Dialogue 2, Jamboard was used to capture notes and for participants to directly contribute comments.

Surveys

Survey distribution and data collection was carried out using the online platform Qualtrics. All survey data was collected by Imperial College London and anonymous data shared with the UWI for analysis.

Data storage and sharing

Data was stored and managed by Imperial College London using a secure server. The UWI was a Joint Data Controller for the data provided to this project for LAC and responsible for securely storing and sharing data with Imperial College London and with regional analyst teams. Data will be stored by Imperial College London for 10 years after study completion.

Appendix 8: Climate hazards in LAC[22, 26]

Observed climate-related hazards in LAC and projected changes

To guide informed discussions on the current and potential mental health consequences of current and future climate change in the region, it was vital to ensure a grounding in climate science. The Red Cross Red Crescent Climate Centre (RCCC) conducted a mapping of previous and projected climate hazards across the region to inform the dialogues and research agenda. The following is a summary of the climate-related hazards that LAC have experienced over the last 30 years, and their projected increases due to climate change to approximately 2030.

It is important to emphasise that the data gathered for the past and future periods differ not only in spatial resolution but are also based on different underlying principles. The former are through observations or documented occurrences such as those reported by national weather services and are aggregated at a country-scale. The latter are from climate modelling-based studies and assessment reports published by the Intergovernmental Panel on Climate Change (IPCC), which are at a more granular spatial scale. For the same reason, the categories of climate-related hazards in the past and future periods may at times not be identical but are generally comparable or related. For instance, flooding that is reported as a relevant climate hazard in the last 30 years here is not a direct output of global climate models (GCMs). Instead, the underlying meteorological variable (rainfall) from the simulations of GCMs are used for deriving indices that can be considered as a proxy for potential flooding in a future warming climate.

LAC

The LAC region is one of the most vulnerable regions to climate change. This region has experienced strong droughts, wildfires, floods, landslides, storms and extreme temperatures in recent years, shown by the international disaster database EM-DAT. Note that the limited number of extreme events recorded (e.g., extreme temperatures) may be due to the missing records in the EM-DAT data for various reasons, such as poor or limited network of satellite and surface observations over the region, or limited reporting from national meteorological services. In addition, the spatial aggregation of the data to country-scales makes it difficult to map the occurrence of the past hazards to human settlements. The total number of people affected as reported in EM-DAT therefore may not always be truly representative of the frequency of occurrences of the underlying climate hazard. For instance, a country in Latin America and the Caribbean may have experienced frequent heatwaves, but if these events were not recorded or in uninhabited regions, the total number of occurrences discussed below may be an underreporting.

<u>Previous Droughts</u>

Droughts and wildfires occurred most frequently in Brazil consequently affecting[vi] the highest number of people. A total of 15 such events affected an average of 7.6 million people between 1991-2022 (Figure 1, top). During this same period, 1.4 million, 1.2 million and 1.1 million people were on average also affected in Haiti, Mexico and Guatemala respectively as a result of 5-7 such events in these countries.

[vi] Note, EM-DAT definition of "total affected" accounts for "the total of injured, affected, and homeless people".

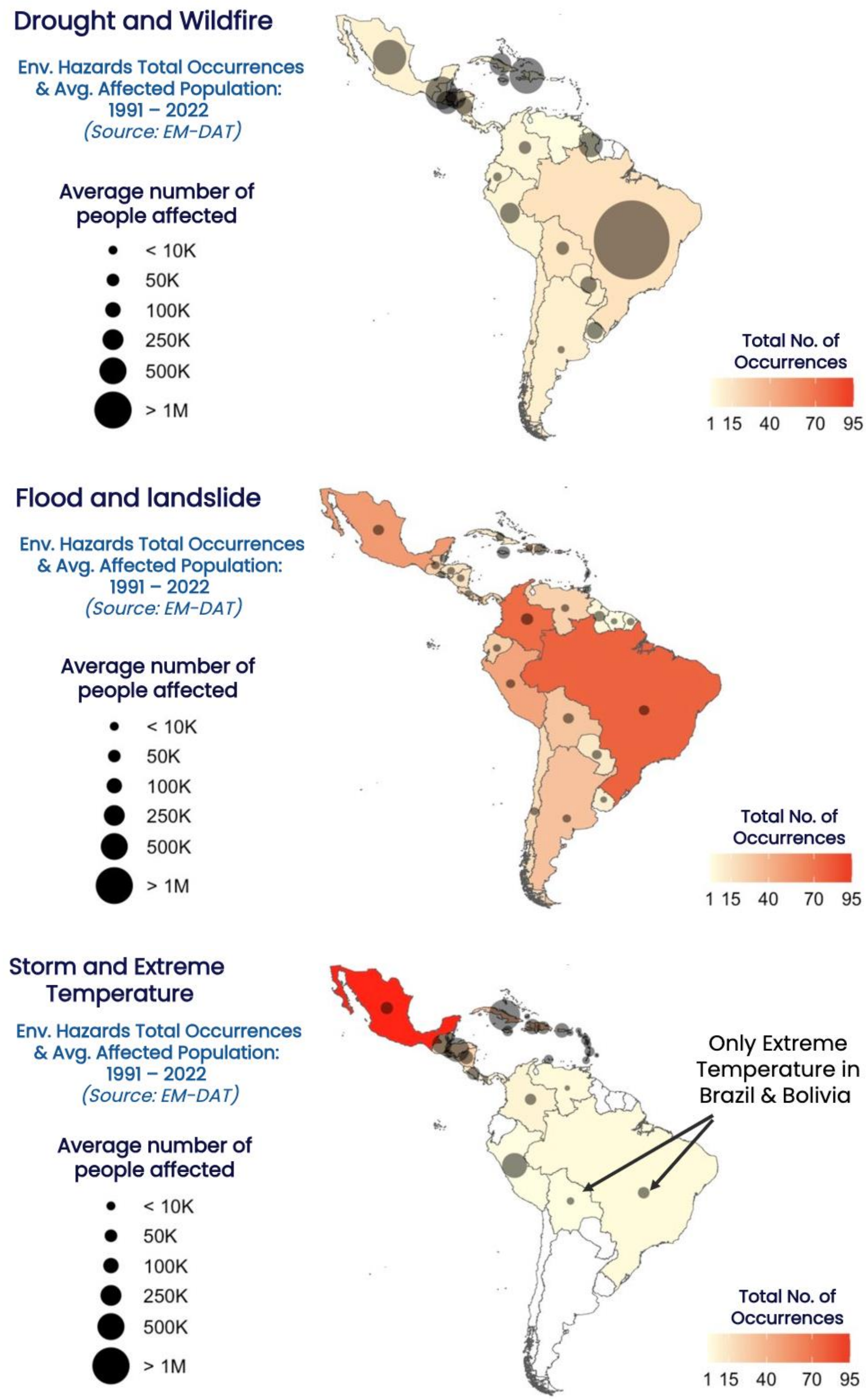


Figure 1. Number of reported droughts and wildfires (top), flood and landslides (middle) and storms and extreme temperatures (bottom) between 1991-2022. Red legend depicts the total number of occurrences of each category of hazard between 1991-2022, the black circle size is representative of the average affected population in that period.

Previous Floods

Floods and landslides were highest in Brazil, followed by Colombia, with 78 and 74 hazards recorded between 1991-2022 (Figure 1, middle) in the two countries respectively. Countries with the highest number of affected people included the Dominican Republic where 167,000 people were affected by 19 such hydrological events. Similarly, Colombia saw approximately 155,000 people affected by 74 such events within this period.

Previous Extreme (Hot) Temperatures and Storms

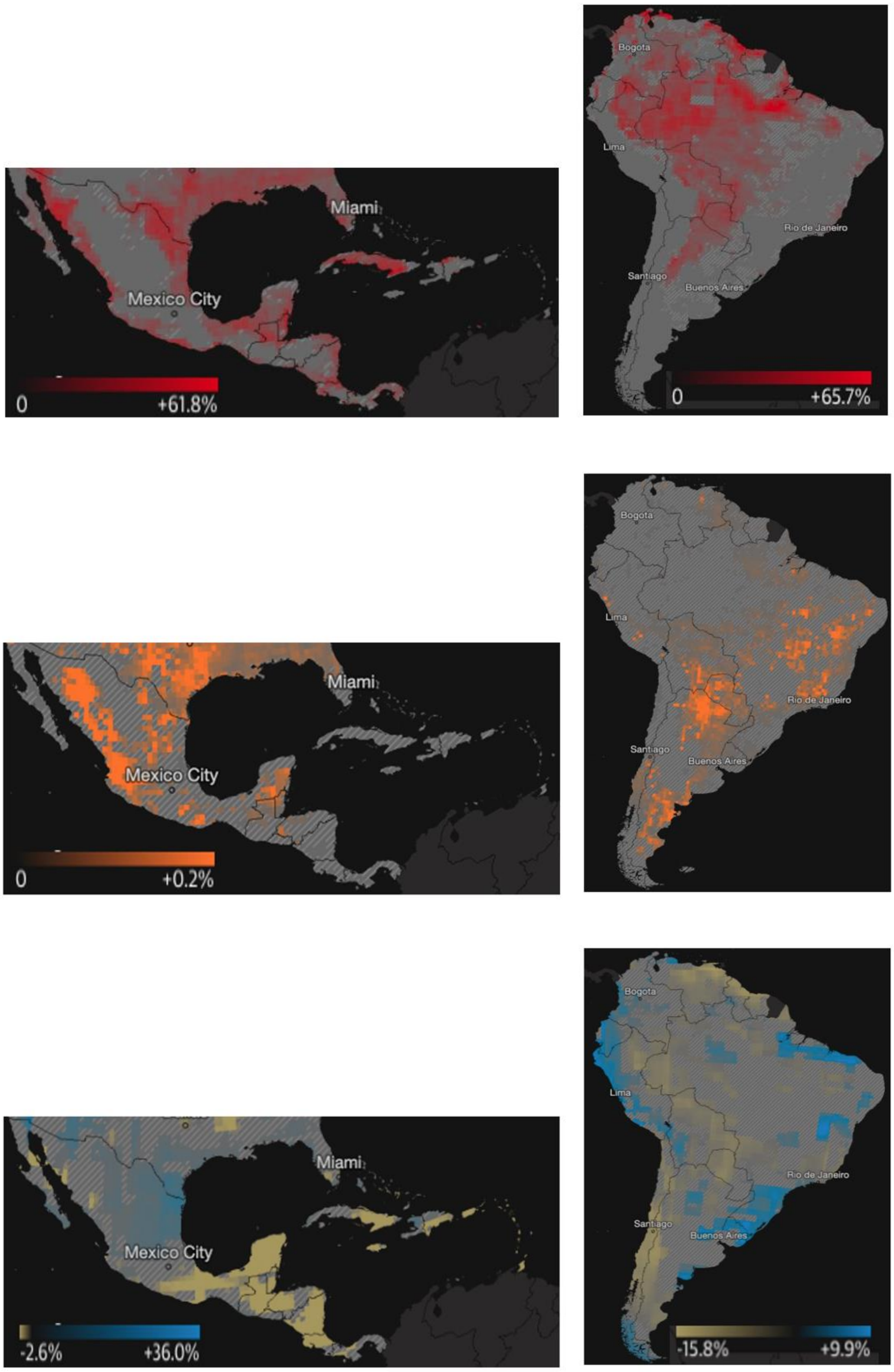
The number of storms and extreme temperatures were particularly high in Mexico, with 92 occurrences of such events affecting near 94,000 people between 1991-2022 (Figure 1, bottom). Cuba and Honduras are countries with population most affected by these meteorological events. While about 970,000 people were affected on average by 31 such events in Cuba, Honduras recorded about 545,000 people affected by 19 events during the same period. Only data on extreme temperatures (not storms) were reported for Brazil and Bolivia in the EM-DAT.

Future Projections

Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the Coupled Model Intercomparison Project - Phase VI (CMIP6) multi-model GCMs ensemble provided in the IPCC, 2021[2]. While not the worst-case emissions scenario, SSP2-4.5 assumes that the 2015 Paris agreement commitments are not achieved and 2°C global warming is not avoided. GCMs have varying climate sensitivity and generally also differ in their representation of earth system processes. While their projections in the intensity of a particular hazard in the future may therefore differ despite a persistent signal in historical observed records, the direction of the signal (i.e., a positive or negative change of a climate hazard) are generally more robust across the GCMs.

- Mean and extreme temperatures: Projections showing the average change in the proportion of land exposed to heatwaves in 2030 relative to 1986-2005 present the tropical regions of South America, Central America and Cuba, and the coastal regions of Mexico are most at risk for such hazards (medium confidence) (Figure 2a). However, the grey stripes over all Figure 2 models indicate there is insufficient agreement across climate models.
- Droughts and wildfires: These projections also show the proportions of land most likely to be exposed to an increase of wildfires are in North Argentina, the more populous regions of Brazil and areas of Mexico (high confidence) (Figure 2b).
- Heavy precipitation events and flooding: The intensity and frequency of extreme precipitation and pluvial floods is projected to increase (medium confidence) for 2°C of global warming level and above (IPCC, 2021). The projected change in annual total rainfall for 2030 using data from 1986-2005 shows the increase in rainfall will be highest in South America (Figure 2c) (medium to high confidence). However, Central America and the Caribbean Islands are more likely to experience a decline in the average rainfall (low confidence).
- Sea level rise: Relative sea level rise is extremely likely to continue in the oceans around Central and South America, contributing to increased coastal flooding in low-lying areas (high confidence) and shoreline retreat along most sandy coasts (high confidence).
- Tropical cyclones: Models generally project an increase in frequency and intensity of Category 4-5 tropical cyclones in this region (medium confidence). Although the spread (indicating the range of

change across models) is wide, the median increase is projected to be around 3% relative to 1986-2005.



a)

b)

c)

Figure 2. Projected changes in a) land exposed to heatwaves b) land exposed to wildfires and c) total annual rainfall for 2030 compared to 1986-2005 under the middle of the road emissions scenario SSP2-4.5.

Note: Detailed information on the projections and the associated confidence are accessed from the IPCC Working Group I contribution to the Sixth Assessment Report "Climate Change 2021: The Physical Science

Basis" and the associated IPCC WGI Interactive Atlas and regional factsheets: Fact Sheets | Climate Change 2021: The Physical Science Basis (ipcc.ch)EM-DAT - The international disaster database (emdat.be)

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